AI505

Optimization

# Constrained Optimization

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# Constrained Optimization

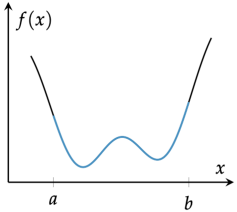
- Minimizing an objective subject to design point restrictions called **constraints**
- A variety of techniques transform constrained optimization problems into unconstrained problems
- New optimization problem statement

$$\begin{aligned} \underset{\boldsymbol{x}}{\text{minimize}} \;\; & f(\boldsymbol{x}) \\ \text{subject to} \;\; & \boldsymbol{x} \in \mathcal{X} \end{aligned}$$

- The set $\mathcal{X} \subset \mathbb{R}$ is called the **feasible set**

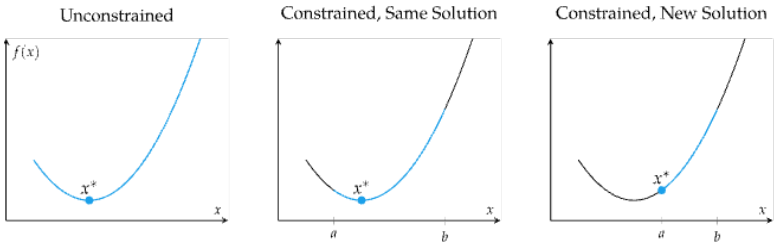
# Constrained Optimization

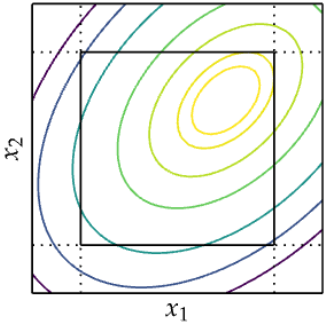
Constraints that bound feasible set can change the optimizer




$$\underset{x}{\text{minimize}}\ f(x)$$

$$\text{subject to } x \in [a, b]$$



# Constraint Types

- Generally, constraints are formulated using two types:
  1. **Equality constraints**: $h(\boldsymbol{x}) = 0$
  2. **Inequality constraints**: $g(\boldsymbol{x}) \leq 0$
- Any optimization problem can be written as

$$\begin{aligned} \underset{\boldsymbol{x}}{\text{minimize}} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \leq 0 \text{ for all } i \text{ in } \{1, \ldots, m\} \\ & h_j(\boldsymbol{x}) = 0 \text{ for all } j \text{ in } \{1, \ldots, \ell\} \end{aligned}$$

$$\begin{aligned} \underset{\boldsymbol{x}}{\text{minimize}} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & \boldsymbol{g}(\boldsymbol{x}) \leq \boldsymbol{0} \\ & \boldsymbol{h}(\boldsymbol{x}) = \boldsymbol{0} \end{aligned}$$

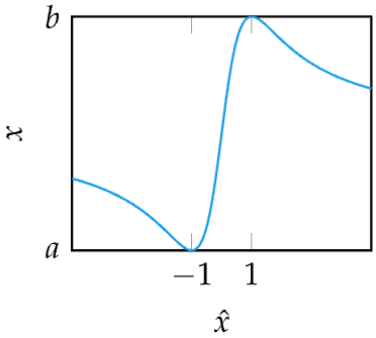
$f$ and the functions $h$ and $g$ are all smooth, real-valued functions on a subset of $\mathbb{R}^n$

# Transformations to Remove Constraints

- If necessary, some problems can be reformulated to incorporate constraints into the objective function

- If $x$ is constrained between $a$ and $b$

$$x = t_{a,b}(\hat{x}) = \frac{b+a}{2} + \frac{b-a}{2}\left(\frac{2\hat{x}}{1+\hat{x}^2}\right)$$

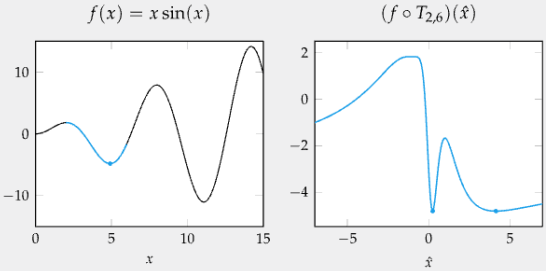


# Transformations to Remove Constraints

## Example

$$\underset{x}{\text{minimize}} \;\; x\sin(x)$$

$$\text{subject to} \;\; 2 \leq x \leq 6$$




$$\underset{\hat{x}}{\text{minimize}} \; t_{2,6}(\hat{x})\sin(t_{2,6}(\hat{x}))$$

$$\underset{\hat{x}}{\text{minimize}} \; \left(4 + 2\left(\frac{2\hat{x}}{1+\hat{x}^2}\right)\right)\sin\left(4 + 2\left(\frac{2\hat{x}}{1+\hat{x}^2}\right)\right)$$

# Transformations to Remove Constraints

## Example

$$\begin{aligned} &\underset{x}{\text{minimize}} \;\; f(\boldsymbol{x}) \\ &\text{subject to} \;\; h(\boldsymbol{x}) = x_1^2 + x_2^2 + \ldots + x_n^2 - 1 = 0 \end{aligned}$$

- Solve for one of the variables to eliminate constraint:

$$x_n = \pm\sqrt{1 - x_1^2 - x_2^2 - \ldots - x_{n-1}^2}$$

- Transformed, unconstrained optimization problem:

$$\underset{x}{\text{minimize}} \;\; \left(\left[x_1, x_2, \ldots, x_{n-1}, \pm\sqrt{1 - x_1^2 - x_2^2 - \ldots - x_{n-1}^2}\right]\right)$$

# Lagrangian Multiplier Method

- With only equality constraints, critical points (local minima, global minima, or saddle points optimal) where gradient of $f$ and the gradient of $h$ are aligned
- The method of Lagrangian Multipliers is used to optimize a function subject to (equality) constraints
- Lagrangian multipliers refer to the variables introduced by the method denoted by $\lambda$

$$\begin{aligned} \underset{\boldsymbol{x}}{\text{minimize}} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & h(\boldsymbol{x}) = 0 \end{aligned}$$

1. Form **Lagrangian function**

$$\mathcal{L}(\boldsymbol{x}, \lambda) = f(\boldsymbol{x}) - \lambda h(\boldsymbol{x})$$

2. Set $\nabla_{\boldsymbol{x}} \mathcal{L}(\boldsymbol{x}, \lambda) = 0$ and $\nabla_{\lambda} \mathcal{L}(\boldsymbol{x}, \lambda) = 0$ to get

$$\nabla f(\boldsymbol{x}) = \lambda \nabla h(\boldsymbol{x}) \qquad h(\boldsymbol{x}) = 0$$

3. solve for $\boldsymbol{x}$ and $\lambda$

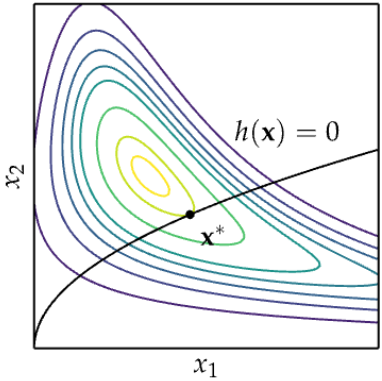
## Example

$$\text{minimize} \quad -\exp\left(-\left(x_1 x_2 - \frac{3}{2}\right)^2 - \left(x_2 - \frac{3}{2}\right)^2\right)$$

$$\text{subject to } x_1 - x_2^2 = 0$$

# Lagrangian Multiplier Method

Intuitively, the method of Lagrange multipliers finds the point $x^*$ where the constraint function is orthogonal to the gradient



# Lagrangian Multiplier Method with Inequality Constraints

$$\begin{aligned} \underset{\boldsymbol{x}}{\text{minimize}} \;\; & f(\boldsymbol{x}) \\ \text{subject to} \;\; & g(\boldsymbol{x}) \leq 0 \end{aligned}$$

- If solution lies at the constraint boundary, the constraint is called **active**, and the Lagrangian condition holds for a non-negative constant $\mu$:

$$\nabla f(\boldsymbol{x}) + \mu \nabla g(\boldsymbol{x}) = 0$$

- If the solution lies within the boundary, the constraint is called **inactive**, and the optimal solution simply lies where

$$\nabla f(\boldsymbol{x}) = 0$$

that is, the Lagrangian condition holds with $\mu = 0$

# Lagrangian Multiplier Method with Inequality Constraints

$$\underset{\boldsymbol{x}}{\text{minimize}}\ f(\boldsymbol{x})$$

$$\text{subject to}\ \ g(\boldsymbol{x}) \leq 0$$

- We create the Lagrangian function such that it goes to $\infty$ outside the feasibility set ($g(\boldsymbol{x}) \not\leq 0$)):

$$\mathcal{L}_\infty(\boldsymbol{x}) = f(\boldsymbol{x}) + \infty(g(\boldsymbol{x}) > 0)$$

impractical: discontinuous and nondifferentiable.

- Instead, for $\mu \geq 0$:

$$\mathcal{L}(\boldsymbol{x}, \mu \geq 0) = f(\boldsymbol{x}) + \mu g(\boldsymbol{x})$$

$$\mathcal{L}_\infty(\boldsymbol{x}) = \underset{\mu \geq 0}{\text{maximize}}\ \mathcal{L}(\boldsymbol{x}, \mu)$$

for $\boldsymbol{x}$ infeasible, $\mathcal{L}_\infty(\boldsymbol{x}) = \infty$; for $\boldsymbol{x}$ feasible, $\mathcal{L}_\infty(\boldsymbol{x}) = f(\boldsymbol{x})$

- The new optimization problem becomes

$$\underset{\boldsymbol{x}}{\text{minimize}}\ \underset{\mu \geq 0}{\text{maximize}}\ \mathcal{L}(\boldsymbol{x}, \mu)$$

This is called the **primal problem**

# Necessary Conditions – KKT Conditions

$$\begin{aligned} \underset{\boldsymbol{x}}{\text{minimize}} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & \boldsymbol{g}(\boldsymbol{x}) \leq \boldsymbol{0} \\ & \boldsymbol{h}(\boldsymbol{x}) = \boldsymbol{0} \end{aligned}$$

Any critical point $\boldsymbol{x}^*$ must satisfy the **Karush-Kuhn-Tucker conditions**

1. primal feasibility: $\boldsymbol{g}(\boldsymbol{x}^*) \leq 0$ and $\boldsymbol{h}(\boldsymbol{x}^*) = 0$
2. dual feasibility: penalization is towards feasibility $\boldsymbol{\mu} \geq 0$
3. complementary slackness: either $\mu_i$ or $g_i(\boldsymbol{x}^*)$ is zero.

$$\boldsymbol{\mu} \cdot \boldsymbol{g}(\boldsymbol{x}) = 0 \implies \mu_i g_i(\boldsymbol{x}^*) = 0, \text{ for } i = 1, \ldots, m.$$

4. stationarity: objective function tangent to each active constr.

$$\nabla f(\boldsymbol{x}^*) + \sum_i \mu_i \nabla g_i(\boldsymbol{x}^*) + \sum_j \lambda_j \nabla h_j(\boldsymbol{x}^*) = 0$$

These are the First Order Necessary Conditions (FONC) for Constrained Optimization

# Necessary Conditions – KKT Conditions

In vector form:

$$\begin{cases} \nabla f(\boldsymbol{x}^*) + \boldsymbol{\mu} \cdot \nabla \boldsymbol{g}(\boldsymbol{x}^*) + \boldsymbol{\lambda} \cdot \nabla \boldsymbol{h}(\boldsymbol{x}^*) = \boldsymbol{0} \\ \boldsymbol{\mu} \cdot \boldsymbol{g}(\boldsymbol{x}^*) = 0 \\ \boldsymbol{g}(\boldsymbol{x}^*) \leq \boldsymbol{0}, \ \boldsymbol{h}(\boldsymbol{x}^*) = \boldsymbol{0} \\ \boldsymbol{\mu} \geq \boldsymbol{0} \end{cases}$$

Particular cases:

- $f$ concave, $\boldsymbol{g}$ convex: then KKT are also sufficient
- $\boldsymbol{x}^*$ interior point
- Patological cases where they do not hold (formal expressions of the conditions include assumptions to avoid these cases).

# Duality

- Generalized Lagrangian Function:

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{\mu}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) + \sum_i \mu_i g_i(\boldsymbol{x}) + \sum_j \lambda_j h_j(\boldsymbol{x})$$

- the **primal form** of the optimization problem

$$\underset{\boldsymbol{x}}{\text{minimize}} \; \underset{\boldsymbol{\mu} \geq 0, \boldsymbol{\lambda}}{\text{maximize}} \; \mathcal{L}(\boldsymbol{x}, \boldsymbol{\mu}.\boldsymbol{\lambda})$$

- Reversing the order of operations leads to the **dual form**

$$\underset{\boldsymbol{\mu} \geq 0, \boldsymbol{\lambda}}{\text{maximize}} \; \underset{\boldsymbol{x}}{\text{minimize}} \; \mathcal{L}(\boldsymbol{x}, \boldsymbol{\mu}, \boldsymbol{\lambda})$$

- What is the relationship between the two? (See next slide)
- In some cases, the dual problem is easier to solve computationally than the original problem.
  In other cases, the dual can be used to obtain easily to gain information on the optimal value of the objective for the primal problem.
  The dual has also been used to design algorithms for solving the primal problem.

# Min-Max Inequality

**Theorem (Max-min inequality)**

*For any function* $f : Z \times W \to \mathbb{R}$ ,

$$\sup_{z \in Z} \inf_{w \in W} f(z, w) \leq \inf_{w \in W} \sup_{z \in Z} f(z, w) .$$

<u>Proof</u>

For all $z \in Z$, $\inf_{w \in W} f(z, w) \leq f(z, w)$ for all $w \in W$ by definition of infimum being a lower bound.

For all $w \in W$, $f(z, w) \leq \sup_{z \in Z} f(z, w)$ for all $z \in Z$ by definition of supremum being an upper bound.

Let $g(z) \triangleq \inf_{w \in W} f(z, w)$. Let $h(w) \triangleq \sup_{z \in Z} f(z, w)$. Then, for all $z \in Z$ and $w \in W$, $g(z) \leq h(w)$

$\sup_{z \in Z} g(z) \leq h(w)$ holds for all $w \in W$ because the supremum is the least upper bound.

$\sup_{z \in Z} g(z) \leq \inf_{w \in W} h(w)$ because the infimum is the greatest lower bound

$\square$

# Min-Max Inequality and Duality

- When $f$, $W$, and $Z$ are convex the inequality becomes equality and we have a strong max-min property (or a saddle-point property).
- For us:

$$\underset{\boldsymbol{\mu}\geq 0,\boldsymbol{\lambda}}{\text{maximize}}\ \underset{\boldsymbol{x}}{\text{minimize}}\ \mathcal{L}(\boldsymbol{x},\boldsymbol{\mu},\boldsymbol{\lambda}) \leq \underset{\boldsymbol{x}}{\text{minimize}}\ \underset{\boldsymbol{\mu}\geq 0,\boldsymbol{\lambda}}{\text{maximize}}\ \mathcal{L}(\boldsymbol{x},\boldsymbol{\mu}.\boldsymbol{\lambda})$$

- Therefore, the solution to the dual problem $d^*$ is a **lower bound** to the primal solution $p^*$
- The inner part of the dual problem can be used to define the **dual function** or **dual objective**

$$\mathcal{D}(\boldsymbol{\mu}\geq 0,\boldsymbol{\lambda}) = \underset{\boldsymbol{x}}{\text{minimize}}\ \mathcal{L}(\boldsymbol{x},\boldsymbol{\mu},\boldsymbol{\lambda})$$

# Duality

- The dual function is concave. Gradient ascent on a concave function always converges to the global maximum.
- **Dual Problem**: $\max \mathcal{D}(\lambda)$ subject to $\lambda \geq 0$
- Optimizing the dual problem is easy whenever minimizing the Lagrangian with respect to $x$ is easy.
- For any $\boldsymbol{\mu} \geq \mathbf{0}$ and any $\boldsymbol{\lambda}$, we have

$$\mathcal{D}(\boldsymbol{\mu} \geq \mathbf{0}, \boldsymbol{\lambda}) \leq p^*$$

- The difference between dual and primal solutions $d^*$ and $p^*$ is called the **duality gap**
- Showing zero-duality gap is a "certificate" of optimality

# Penalty methods

- Penalty methods are a way of reformulating a constrained optimization problem as an unconstrained problem by penalizing the objective function value when constraints are violated

**Example**

$$\begin{aligned} \underset{\boldsymbol{x}}{\text{minimize}} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & \boldsymbol{g}(\boldsymbol{x}) \leq \boldsymbol{0} \\ & \boldsymbol{h}(\boldsymbol{x}) = \boldsymbol{0} \end{aligned}$$

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & f(\boldsymbol{x}) + \rho \cdot p_{count}(\boldsymbol{x}) \\ \text{s.t.} \quad & p_{count}(\boldsymbol{x}) = \sum_i (g_i(\boldsymbol{x}) > 0) + \sum_j (h_j(\boldsymbol{x}) \neq 0) \end{aligned}$$

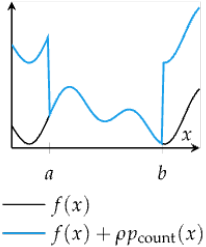
# Penalty Methods

**Procedure** penalty_method;
**Input:** $f, p, \boldsymbol{x}, k_{max}; \rho = 1, \gamma = 2$
**Output:** $\boldsymbol{x}$ solution
**for** $k$ in $1, \ldots, k_{max}$ **do**
  $\boldsymbol{x} \leftarrow \text{minimize}_{\boldsymbol{x}}\{f(\boldsymbol{x}) + \rho \cdot p(\boldsymbol{x})\}$;
  $\rho \leftarrow \rho \cdot \gamma$;
  **if** $p(\boldsymbol{x}) = 0$ **then**
    return $\boldsymbol{x}$;
return $\boldsymbol{x}$;

# Penalty methods

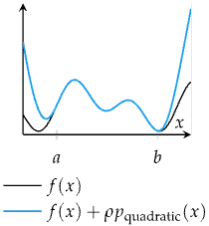
- Count penalty:

$$p_{count}(\mathbf{x}) = \sum_i (g_i(\mathbf{x}) > 0) + \sum_j (h_j(\mathbf{x}) \neq 0)$$



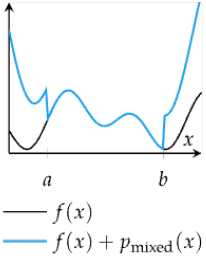

- Quadratic penalty:

$$p_{quadratic}(\mathbf{x}) = \sum_i \max(g_i(\mathbf{x}), 0)^2 + \sum_j h_j(\mathbf{x})^2$$




- Mixed Penalty:

$$p_{mixed}(\mathbf{x}) = \rho_1 p_{count}(\mathbf{x}) + \rho_2 p_{quadratic}(\mathbf{x})$$



# Augmented Lagrange Method

- Adaptation of penalty method for equality constraints

$$p_{Lagrangian}(\boldsymbol{x}) \stackrel{def}{=} \frac{1}{2}\rho \sum_i (h_i(\boldsymbol{x}))^2 - \sum_i \lambda_i h_i(\boldsymbol{x})$$

**Procedure** augmented_lagrange_method;
**Input:** $f, h, \boldsymbol{x}, k_{max}; \rho = 1, \gamma = 2)$
$\boldsymbol{\lambda} \leftarrow \boldsymbol{0}$;
**for** $k$ in $1, \ldots, k_{max}$ **do**
  $p \leftarrow (x \mapsto \rho/2 \cdot \sum_i (h_i(\boldsymbol{x})^2) - \boldsymbol{\lambda} \cdot h(\boldsymbol{x}))$;
  $\boldsymbol{x} \leftarrow \text{minimize}_{\boldsymbol{x}}\{f(\boldsymbol{x}) + p(\boldsymbol{x})\}$;
  $\boldsymbol{\lambda} \leftarrow \boldsymbol{\lambda} - \rho \cdot \boldsymbol{h}(\boldsymbol{x})$;
  $\rho \leftarrow \rho \cdot \gamma$;
return $\boldsymbol{x}$;

- $\boldsymbol{\lambda}$ converges towards the Lagrangian multiplier

# Interior Point Methods

$$\begin{aligned} \underset{\boldsymbol{x}}{\text{minimize}} \;\; & f(\boldsymbol{x}) \\ \text{subject to} \;\; & \boldsymbol{g}(\boldsymbol{x}) \leq \boldsymbol{0} \end{aligned}$$

- Also called **barrier methods**, interior point methods ensure that each step is feasible
- This allows premature termination to return a nearly optimal, feasible point
- Barrier functions are implemented similar to penalties but must meet the following conditions:
  1. Continuous
  2. Non-negative in the feasible region
  3. Approach infinity as x approaches boundary

# Interior Point Methods

- Inverse Barrier:

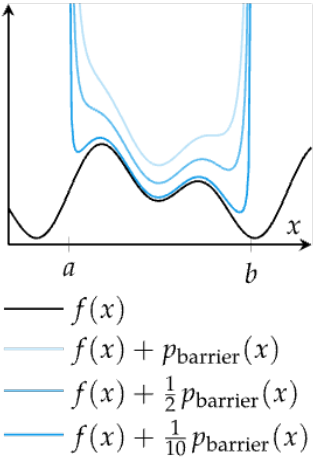
$$p_{barrier}(\mathbf{x}) = -\sum_i \frac{1}{g_i(\mathbf{x})}$$

- Log Barrier:

$$p_{barrier}(\mathbf{x}) = -\sum_i \begin{cases} \log(-g_i(\mathbf{x})) & \text{if } g_i(\mathbf{x}) \geq -1 \\ 0 & \text{otherwise} \end{cases}$$

New optimization problem:

$$\underset{\mathbf{x}}{\text{minimize}}\ f(\mathbf{x}) + \frac{1}{\rho} p_{barrier}(\mathbf{x})$$



# Interior Point Methods

**Procedure** interior_point_method;
**Input:** $f, p, \boldsymbol{x}; \rho = 1, \gamma = 2, \epsilon = 0.001$
$\Delta \leftarrow \infty$;
**while** $\Delta > \epsilon$ **do**
  $\boldsymbol{x}' \leftarrow \text{minimize}_{\boldsymbol{x}}\{f(\boldsymbol{x}) + p(\boldsymbol{x})/\rho\}$;
  $\Delta \leftarrow \|\boldsymbol{x}' - \boldsymbol{x}\|$;
  $\boldsymbol{x} \leftarrow \boldsymbol{x}'$;
  $\rho \leftarrow \rho \cdot \gamma$;
return $\boldsymbol{x}$;

- Line searches $f(\boldsymbol{x} + \alpha \boldsymbol{d})$ are constrained to the interval $\alpha = [0, \alpha_u]$, where $\alpha_u$ is the step to the nearest boundary.
  In practice, $\alpha_u$ is chosen such that $\boldsymbol{x} + \alpha \boldsymbol{d}$ is just inside the boundary to avoid the boundary singularity.
- Needs an initial **feasible** solutions. Typically, found by solving:

$$\underset{\boldsymbol{x}}{\text{minimize}}\ p_{quadratic}(\boldsymbol{x})$$

# Summary

- Constraints are requirements on the design points that a solution must satisfy
- Some constraints can be transformed or substituted into the problem to result in an unconstrained optimization problem
- Analytical methods using Lagrange multipliers yield the generalized Lagrangian and the necessary conditions for optimality under constraints
- A constrained optimization problem has a dual problem formulation that is easier to solve and whose solution is a lower bound of the solution to the original problem
- Penalty methods penalize infeasible solutions and often provide gradient information to the optimizer to guide infeasible points toward feasibility
- Interior point methods maintain feasibility but use barrier functions to avoid leaving the feasible set